

The 13th International Conference on Emerging Ubiquitous Systems and Pervasive Networks
(EUSPN 2022)
October 26-28, 2022, Leuven, Belgium

Privacy Threat Analysis for connected and autonomous vehicles

Badreddine Chah^{*,a}, Alexandre Lombard^a, Anis Bkakria^b, Reda Yaich^b, Abdeljalil
Abbas-Turki^a, Stéphane Galland^a

^aCIAD UMR 7533, Univ. Bourgogne-Franche-Comté, UTBM, F-90010 Belfort, France

^bIRT SystemX, Palaiseau 91120, France

Abstract

Connected and autonomous vehicles produce, store and communicate a large amount of personal data (the route taken, the stop points, home and work addresses, etc.). The development of this type of vehicles brings the opportunity to offer new services to road users, with more software and hardware components on the vehicle. Many of these components suffer from weaknesses that can be exploited. The issue is that a single vulnerability in one element of the system will threaten the privacy of the vehicle's users (The weakest link principle of IT security). In this paper, we present a privacy threat analysis on the general architecture of connected and autonomous vehicles, to point out privacy risks according to formal privacy requirements. We present a use case modularization and its analysis that helps us to discern the privacy requirements of this use case, which can be enabled by manufacturers. To meet the previous objectives, we follow a *LINDDUN* methodology of privacy threat analysis.

© 2022 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the Conference Program Chairs

Keywords: Privacy, Security, Threat Analysis, Connected and Autonomous Vehicle, Privacy engineering framework, Cyber-Physical Systems (CPS).

1. Introduction

Each Connected and Autonomous Vehicle (CAV) uses a number of sensors and Electronic Control Units (ECUs) to collect data on the vehicle's behavior and environment, and to control the vehicle's functionality. Depending on the use cases (also called applications or services), the ECUs collaborate with each other and with external entities by exchanging information related to the vehicle and/or its occupants. With the growth of the provided services to the

* Corresponding author. Tel.: +33-384-583-319 ; fax: +33-384-583-342.

E-mail address: badreddine.chah@utbm.fr

vehicle's occupants, a high-performance computer (HPC¹) connected to the ECUs are needed [29]. We make up two types of networks: Onboard network and Outboard networks (V2X), that we will describe in detail in the following sections. V2X is an acronym for Vehicle to Everything, and it is a vehicular communication system that will allow vehicles to exchange information with each other (V2V), with the infrastructure (V2I), with pedestrians (V2P) and so on. By 2025, in lower levels of autonomy, vehicles will produce 25GB of data per hour and up to 500GB if fully autonomous [15]. Nevertheless, the collection of personal data requires a strict framework to prevent possible abuses. Currently, at the European Union (EU) level, it is the General Data Protection Regulation (GDPR) [5] that legally regulates the use of data and protects the users' privacy and personal information, providing a single legal framework for professionals. To meet the GDPR legal requirements, car manufacturers and their suppliers have to ensure that their developed systems will preserve the privacy of the users. To assist manufacturers in their work, standards have been presented, which improve security and privacy in CAVs[1, 2]. In addition, this work can help to highlight potential attacks before their occurrence. With a focus on data-driven privacy, in order to comply with the GDPR requirements.

In the literature, there is a growing research interest in studying privacy in CAV. Some related work explain the privacy interests and privacy expectations in CAVs (such as in [10]), but the authors are not projected into real-world scenarios, and they do not analyze the privacy threats according to vehicle's architecture. Others are focused on preserving privacy in each specific scenario and use case (application), such as the papers dealing with the Smart Parking Navigation use case [11]. Others are interested in studying the main security and privacy challenges for connected and autonomous vehicles, with a slight bias towards privacy [4, 22, 20]. They focus mostly on security threat analysis. Nevertheless, they do not present any privacy risk analysis, by focusing on security and overlooking the privacy requirements of the GDPR. To the best of our knowledge, no attempt has been made to analyze the privacy threats in CAVs field that follows a privacy threat analysis methodology.

In this paper we study how the LINDDUN privacy threat analysis framework [6] can be used for the elicitation and analysis of privacy requirements of CAV system, while respecting the privacy properties set by the GDPR (which are: Unlinkability, Anonymity, Plausible deniability, Undetectability, and Confidentiality). We use a modeling system process to deal with the threats to personal data flow in Onboard/Outboard networks, by analyzing the type of exchanged data, attack vectors, and the various privacy threats in CAVs. By consequence, it can help to highlight potential attacks before their occurrence. With a focus on data-driven privacy, in order to comply with the GDPR requirements.

This paper is organized as follows. Section 2 introduces in a high-level way the general architecture and the component of the CAV, and defines and classifies some use cases. Then, the LINDDUN methodology is detailed in the Section 3. Section 4 presents an application of the methodology to CAVs systems. Section 5 concludes this work.

2. Overview of CAV components and use cases

CAVs have a great potential to develop several services such as reducing road accidents, improving the quality of life, and enhancing the efficiency of transportation systems. Their communication ability is an evolving technology that promises a massive advance in the field of intelligent transportation systems (ITS). These vehicles are equipped with ECUs which communicate, firstly with other computers through the Onboard vehicle network, and secondly with other external entities through the outboard vehicle network. Even if the architecture differs from one manufacturer to another, the different critical elements that are subject to vulnerable attack vectors can be classified into two parts (see Fig. 1): Onboard vehicle network and outboard vehicle network. **The Onboard vehicle network** is composed of two parties: First, It is the automotive bus systems (i.e. CAN, Ethernet, Flex-Ray, LIN, MOST, etc.), where each subnetwork is dedicated to a specific service that can be linked by a gateway. Second, it is the multiple vehicular ports, like OBD and USB ports, that give physical access to the vehicle's systems such as ECUs. **The outboard vehicle network** includes the elements outside the vehicle. It is based on known technologies to transfer sensitive data: Dedicated Short Range Communications (DSRC)/Wireless Access in Vehicular Environments (WAVE)/WiFi, cellular networks (LTE, 4G and 5G etc.), Bluetooth, GSM, GPS and so on [25]. The data collected/transmitted by these elements is used for providing services that we will be generalized and classified in the following paragraph, according to the data flows.

¹ <https://www.continental-automotive.com/Passenger-Cars/Autonomous-Mobility/Enablers/AD-High-Performance-Computer>

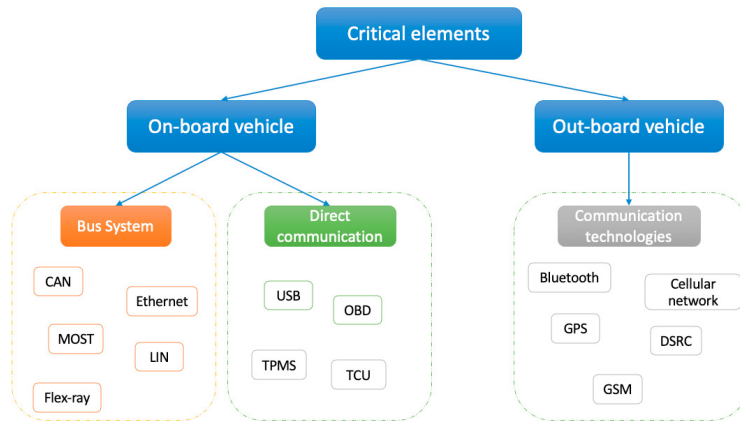


Fig. 1. Taxonomy of critical elements in the CAV architecture

CAVs use cases. The current development of these vehicles brings the opportunity to offer new services to road users. Thus, through the use of data collected and transmitted by other vehicles, it is imaginable that manufacturers could offer new services related to traffic prediction and management, to the enhancement of the driving experience of users (e.g. in-vehicle entertainment), and to the optimization of the vehicle's use (e.g. energy saving and incivility monitoring for shared car). According to the state of the art [7, 8], we classify services into three types:

- First, *IN-IN category* means that the data collected by the vehicle remains in the vehicle. Data are analyzed, processed and stored in the vehicle, like in the case of eco-driving aid system, active wellness², and driver attention assistance³.
- Second, *IN-OUT category* which includes use cases where the data collected in the vehicle are communicated to external entities, as for the following use cases: V2X broadcast messages (CAM) [13], pay-as-you-go insurance⁴, vehicle diagnosis⁵, enhanced e-call⁶, car sharing⁷, and so on.
- Third, *IN-OUT-IN category* includes vehicle's applications that send data to an external entity (e.g. cloud) for processing, analysis and collection, then, waits to receive the results of these operations. IN-OUT-IN applications include, location & navigation Service⁸, cooperative adaptive cruise control⁹, secure automated valet parking [11], and so on.

Potential issues regarding privacy and corresponding regulations, are due to the fact that CAVs are connected to the outside world. The data captured by the vehicle's ECUs can be collected/processed/analysed Onboard the vehicle through ports (e.g. USB port) or during communication to an outsourced service provider. Let us give the example of the case of the shared vehicle service. The autonomous vehicle recognizes each passenger by calling him by his name, and puts his preferred playlist and usual destinations. This data is collected by the vehicle's HPC, through sensors, telematics boxes, ECUs or mobile applications. These sensitive data are also processed inside the vehicle or exported to a third party that is outside the vehicle (cloud service). CAV disclose too much personal information, which threatens the user's privacy in a direct way. Thus, the privacy risk must be analyzed to predict the threat to this data before it is exploited. In the next section, the methodology used for this purpose is presented.

² <https://www.faurecia.com/en/newsroom/active-wellness-car-seat-wins-if-design-award>

³ <https://www.axios.com/2019/08/07/volvo-study-self-driving-cars-fuel-efficiency>

⁴ <https://www.theaa.com/car-insurance/advice/pay-as-you-go-car-insurance>

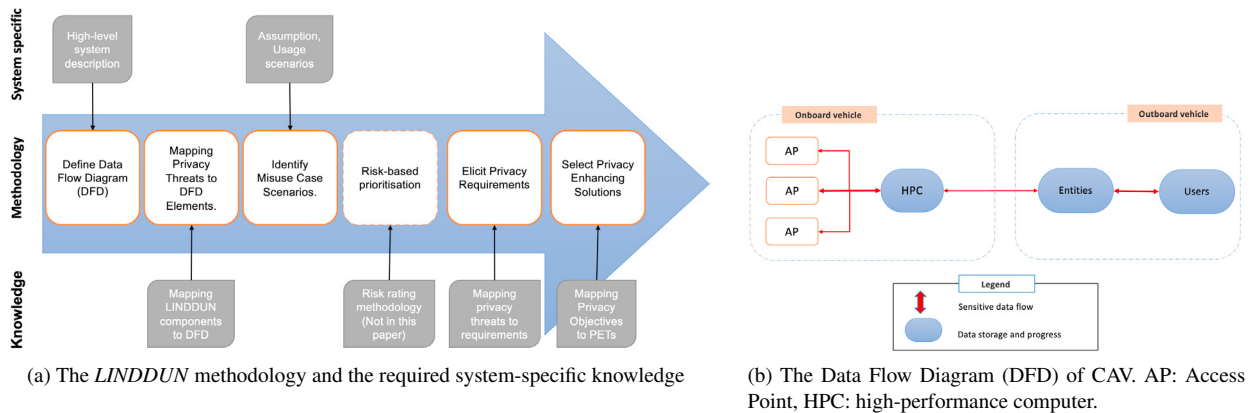
⁵ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6603671/>

⁶ https://www.rohde-schwarz.com/fr/a-propos-de/reportages-techniques/ecall_251205.html

⁷ <https://www.share-now.com/de/en/car-sharing-advantages/>

⁸ <https://www.geoflex.xyz/industry/high-accuracy-positioning-for-navigation-adas-and-autonomous-driving/>

⁹ https://en.wikipedia.org/wiki/Cooperative_Adaptive_Cruise_Control

Fig. 2. Overview of the *LINDDUN* methodology and the CAV DFD diagram

3. Privacy threat methodology (*LINDDUN*)

In [6], Deng et al. present the *LINDDUN* methodology, which is a comprehensive framework to model privacy threats in systems. Other methodologies of privacy threat analysis such as the well-known Privacy Impact Assessment (PIA), also called privacy risk assessment, has a similar process [27, 28]. The *LINDDUN* has been chosen in this work because this methodology defines formal privacy requirements to identify the necessary protections in the form of Privacy Enhancing Technologies (PETs). Moreover, it is a generic methodology [30], that identify the privacy threats, in CAV, in order to respect the privacy properties set by the RGPD (Unlinkability, Anonymity, Plausible deniability, Undetectability, and Confidentiality).

LINDDUN presents the threat modeling process depicted in Fig.2a, which consists of six high-level steps. *Step1: define Data Flow Diagram (DFD)*. DFD is created based on the high-level system description. In our case of study, the analyzed CAV is decomposed into relevant logical and structural elements, and for each of these parts the corresponding threats are reviewed. *Step 2: mapping privacy threats to DFD Elements*. Each element of the DFD is subject to certain privacy threats. Note that the nature of the potential threat to privacy is determined by the type of DFD element. *Step 3: Identify misuse case scenarios*. We start by eliciting privacy threats by using threats trees then the misuse cases. In this step, we answer why it is important to protect users' private data that have been collected from vehicles. Then, we show how privacy threats affect DFD elements. *Step 4: risk-based prioritization*. This step assesses the severity of all the previous identified attack risks and makes an informed decision about what to do about them. *Step 5: elicit privacy requirements*. This defines the privacy objectives that need to be fulfilled. *Step 6: select privacy enhancing solutions*. This last step maps the privacy objectives to the PETs available in the state of the art. The last two steps, i.e. steps 5 and 6, comprise so-called "white hat" activities.

Let us draw the reader's attention to the fact that the scope of the paper is limited to steps 1, 2, 3 and 5 of the methodology. More precisely, this paper focuses on the steps that carry out a threat analysis for privacy and it will neither judge the priority of threats (step 4) nor detail practical solutions to them (step 6). Only a sketch of solutions is given for the step 6. With this in mind, the focus in the next sections is put on the use of the four steps of the methodology for analyzing the privacy threats in the case of CAV.

4. Application of *LINDDUN* to CAV privacy analysis

In order to determine the threat to privacy posed by CAV, we take the following steps:

Step 1: Define data flow diagram (DFD). In this step, we discern the high level of the CAV system and scenarios that can be adapted to each use case and each CAV architecture described in the overview given in section 2. The stakeholders in this scenario are (see Tab.1): Users, Access Points (e.g. OBD II port, telematic control unit (TCU), multimedia, TPMS (type pressure monitoring system), etc.), HPC, and Entities (e.g. service provider (SP), other vehicles and Road Side Units (RSUs)). Fig. 2b shows a general architecture of the data flow Onboard/Outboard CAV

Table 1. DFD elements in the CAVs

DFD Elements Types	DFD Elements Members
Access points	OBD port, multimedia(USB), TCU, TPMS , Electronic vehicles charging (EVC) , etc.
Users	Passenger, Owner
Data flow	AP-AP and AP-HPC through Onboard CAV network (i.e. CAN, ethernet, flex-ray, LIN, etc.), HPC - Entities and HPC - Users and Entities - Users through outboard CAV network (cellular network, WiFi, multimedia (Mult) (Bluetooth), GSM network, etc.)
Data store	Entities data base (i.e. service provider DB, services DB, mobile phone DB, etc.) HPC data base (also called CAV data base)
Process	Entities (i.e. service provider (Prov), services (Ser), mobile phone (Mobl), etc.) HPC

network, which is also the data flow diagram (DFD). The DFD is classified into relevant components (both logical and structural), and for each of its parts, the corresponding threats are analyzed throughout the step 3. The DFD contains the following elements: (1) *Users* are considered to be vehicle's occupants/owners who wish to preserve their personal data. They may also have a mobile phone that is connected to the car and to the cloud/service provider. (2) *Data flow* are the sensitive data circulating between the following endpoints: (i) Between the APs (access points). (ii) Between one AP and ECUs/HPC. (iii) Between ECUs/HPC and Entities. (iv) Between HPC and Users (i.e. passengers, owners or mobile phone). (v) Between entities and users. (3) *Data store*: in our context, we considered the data stored in HPC, Entities (e.g. cloud) and Users (e.g. user's mobile phone and USB key). (4) *Process*: Data processing by the HPC, by the User's device and by the Entities. For the threat modeling, we consider the endpoints in Fig. 2b as malicious, by keeping in mind that this assumption can be relaxed according to the choice of the User. Indeed, some Entities may be considered as trusted, but this depends on the adversary model considered for each service (use case).

Step 2: Mapping privacy threats to DFD elements. *LINDDUN* is an acronym reflecting the categories of threats, aiming to cover the following ones: *Linkability(L)* is the possibility that an adversary has to associate two or more items of interest (e.g. messages, actions). *Identifiability(I)* is the fact that an adversary can sufficiently identify the subject associated to an item of interest. *Non-repudiation(NR)* means that an adversary can prove that a user knows, has done, or has said something. *Detectability(D)* is that the adversary can sufficiently distinguish whether an item of interest exists or not. *Disclosure of Information(DI)* is the exposure of personal information to individuals who are not supposed to have access to it. *Unawareness of content(U)* where the users are unaware of the information disclosed to the system. *Non-Compliance(N)* occurs when the system does not comply with (data protection) regulations, stated policies and existing user consents (for more details, see the section 4 of [6]). These are the bases for the proposed *LINDDUN* framework.

Following the same reasoning of [6], we select the critical elements that can be exploited by a malicious entity and affect the privacy. Therefore, after choosing the relevant DFD elements of the general CAV architecture in **step 1** (see Tab. 1), we then mapped the *LINDDUN* privacy threats to the DFD element. This mapping is presented in Tab. 2. Each element of DFD is subject to one or more privacy threats. These threats are determined by the type of DFD elements. Note that each of the x-mark in the table has been examined in this step. For the sake of the readability of the paper and due to lack of space, in the following, only some examples of the x-marks (red ones) of this table are detailed.

Step 3: Identify misuse case scenarios. This step is divided into two sub-steps: first, we use a threat tree to identify privacy threats for DFD element types. The tree is a conceptual diagram showing how an asset, or a target, could be attacked (See [16]). Second, we describe the misuse case (related to threat tree) to document the threat scenarios of these elements. The misuse case is a sequence of actions, initiated by a misactor, resulting in loss for a stakeholder. A misactor is the opposite of an actor, it is an entity that initiates misuse cases. Note that each type of DFD element marked in the Tab. 1 should be linked with a threat tree and misuse case.

Fig 3 gives an example of the threat tree related to the linkability of DF. In this figure, two preconditions are extracted from the tree: (1) First, the transmitted data between different APs, AP-HPC Onboard, and HPC/TCU-Entities are linkable. If the transmitted data are available to malicious parties connected Onboard/Outboard CAVs, this leads to the disclosure of either sensitive private data or at least, data that can be linked with other ones (i.e. CAV's occupancy,

Table 2. Mapping privacy threats to DFD elements members(**TP**:Threat privacy, **Enti**: entities, **U**: users), **X**:is to point out the example that we took in this paper which is linkability of users data flow. In practice each of the x-mark in the table must be examined.

TP	U	Enti	Enti DB (DS)	Bus Syst (DF)	Cellular (DF)	GSM (DF)	WiFi (DF)	OBD (AP)	Mult (AP)	TCU (AP)	GPS (AP)	TPMS (AP)
L	X	X	X	X	X	X	X	X	X	X	X	X
I	X	X	X	X	X	X	X	X	X	X		X
NR	X	X	X	X	X	X	X	X	X	X		
D	X	X	X	X	X	X	X	X	X	X	X	
DI	X	X	X	X	X	X	X	X	X	X		
N	X	X	X	X	X	X	X	X	X	X		

Table 3. Mapping privacy threats to DFD elements members and shows some vulnerabilities and attack scenarios Onboard/Outboard CAV.

Critical elements	Attack scenarios	Preconditions (Vulnerabilities)	Privacy threat				
			L	I	NR	D	DI
On-board networks [17,18]	(1) Attacker injects CAN messages/codes via any ECUs connected to the bus. (2) Attacker modifies the path of the sensitive data circulating in the vehicle. (3) Attacker remotely accesses via telematics systems (TCU).	The presence of different domains (CAN, Ethernet, FlexRay, LIN, Most...). The data circulating in the vehicles is not encrypted. Sensitive information accessible via ECUs. Gateway vulnerabilities.	✓	✓	✓	✓	✓
V2X Technologies (Cellular network, WiFi [20], Bluetooth [19], GSM Network[21])	(4) An adversary can capture, analyze, and replay the messages. (5) An adversary could link the messages to a specific vehicle. (6) An adversary can crack a weak password. (7) An adversary can execute arbitrary code on the telematics unit (TCU) and take control of the device. (8) The SS7 vulnerabilities can be exploited to compromise SMS system to intercept text user messages. (9) Spying, SMS Interception & Modification, User Tracking.	Bluetooth vulnerabilities. Vulnerabilities in Cellular Networks/ DSRC. Vulnerabilities in the GSM protocol, such as SS7 for SMS.	✓	✓	✓	✓	✓
OBD II Port [22]	(10) Attacker can edit, add, manipulator codes and framework of the must ECUs in the vehicles. (11) Attacker can modify the hardware by changing or disabling physical sensors, memories and features. (12) An attacker changes circuits of a device to extract sensitive data, such as cryptographic keys.	OBU II give access to all ECUs, even those not required for diagnosis. Any entity can connect a hardware to vehicles.	✓	✓	✓	✓	✓
TCU (Telematics Communication Unit) [22]	(13) Attacker operates physically on the TCU by tampering the device firmware. (14) An attacker could perform remote control by installing remotely his own software on the device.	Access through the telematics server (Cloud) or even physical access, allowing vehicle tracking, telemetry data collection, remote control and additional services.	✓	✓	✓	✓	✓
Multimedia (USB) [22-24]	(15) Reprogramming the USB/CD device's firmware to execute malicious actions. (16) An adversary can exploit the vulnerabilities of USB systems on-board vehicle.	Easy physical access. Connecting to USB port and accessing unencrypted drives on the monitor. They are connected directly to the computer centre. Possibility to access to multiple devices.	✓	✓	✓	✓	✓
TPMS [25,26]	(17) An adversary can identify or track the vehicle from the sensor. (18) For wireless communication channel, messages can be captured, analyze, and replayed.	Anyone can send message. TPMS messages sent in clear text.	✓	✓			
GPS [25,26]	(19) Possibility to discover and control the behaviour and profile of the driver. (20) An adversary relates pseudonymous positions to specific vehicles	GPS Security vulnerabilities (technical anonymisation).	✓			✓	

CAV's owner). (2) Contextual data, such as location data (i.e. the CAV's position) are not protected. These data can be easily linked to obtain sensitive information about a user. This is achieved through the non-anonymization of communication systems, and allows contextual data to be linked to the passenger ID, vehicle serial number, etc. In addition, the infrastructure is probably deployed with insecure systems, which may allow traffic analysis and passive or active attacks against the infrastructure (see examples in Tab.3).

Once we have described the previous threat tree, it is used to apply the misuse case. To do this, we follow the misuse case structure presented in [6] that is based on the model provided by [9]. Then, we list the most pertinent scenario attacks that are compatible with all applications/use cases (see the Tab.2).

Title of misuse case:

Linkability of users data flow Onboard/Outboard CAV networks

- **Summary:** Data flows can be linked to the same passenger/owner, without necessarily revealing the identity of that entities.

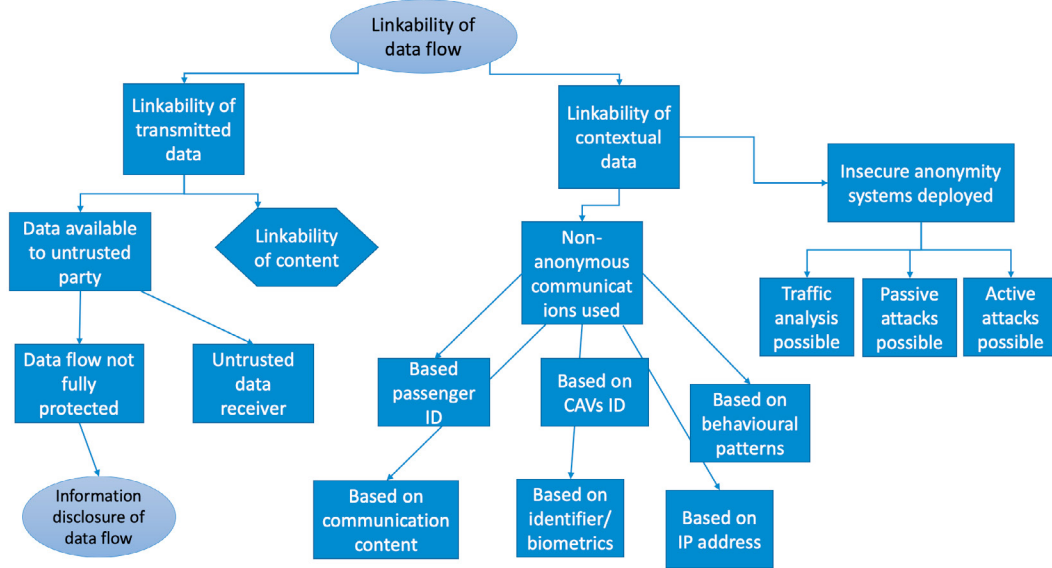


Fig. 3. Threat tree for Linkability privacy threat of a Data Flow onboard/outboard CAVs

- **Assets, stakeholders and threats:** Passenger/Owner: an adversary can obtain the personal information such as location information, users recognition, vehicle material information, their behaviors, etc. The potential damage if this misuse case succeeds are the identity fraud or theft, actual crime, concealment of personal data.
- **Primary adversary:** the adversary is considered in this paper are mechanics, organized crime, thieves and private/public spy agencies.
- **Basic Flow:** In this part, we describe some normal flow of malicious actions in CAVs system that results. There are considered as successful attacks: see the Tab.3 from the attack scenario number 1 to the attack scenario number 9.
- **Trigger:** Physical and remote access
- **Preconditions:** The required preconditions that system must meet for the attack to be feasible are: (i) the use of untrusted and/or unsecured communications to link passenger identifiers, (ii) the different passenger "pseudonyms" are linked to each other based on passengers' interactions with members of the DFD elements (see some examples in section 3).

Step 5 and 6: Elicit privacy requirements and their mapping to the PETs. After presenting misuse cases that are relevant threat scenarios for data flow in CAVs system, its preconditions are based on the threat tree model, and the basic flows are inspired by the system use cases. We can deduce the privacy requirements of the misuse cases that can be used as a basis for stating the confidentiality requirements of the system. In our misuse case above, privacy is required to use and maintain trusted and secure communications Onboard/Outboard.

In Tab.4, we present the basic high-level privacy objective, which maps the threat scenarios to privacy requirements. For the last step of *LINDDUN* methodology, we give the Tab.5 of PETs, by mapping the privacy objectives to the solutions. Note that the PETs is inspired from the systematic literature review proposed in [12].

5. Conclusion and future work

In this work, we have presented an application of the *LINDDUN* method [6] on connected and autonomous vehicle systems, to analyze the threats to privacy. Through the application of the systematic method, we show how *LINDDUN*'s threat categories can help the CAV system designer to discern the privacy properties of the system. The primary contribution is to model a enlever CAV usages, along with a robust and comprehensive threat analysis of these usages. It includes the study of the threats to Onboard/Outboard CAVs network.

Table 4. Mapping privacy threats to requirements to get privacy's goals. P: Processing, DF: Data FLOW, DS:Data Store. The red line is the example considered in this work.

LINDDUN threats	Elementary privacy objectives
Linkability of users DF	(I) Unlinkability of users DF
Linkability of users during P	(II) Unlinkability of users during P
Linkability of users through AP	(III) Unlinkability of users through AP
Identifiability of users in DF	(IV) Anonymity and pseudonymity of users DF
Identifiability of users during P	(V) Anonymity and pseudonymity of users during P
Identifiability of users through AP	(VI) Anonymity and pseudonymity of users through AP
Non-repudiation of users in DF	(VII) Plausible deniability of users DF
Non-repudiation of users during P	(VIII) Plausible deniability of users during P
Non-repudiation of users through AP	(IX) Plausible deniability of users through AP
Detectability of users in DF	(X) Undetectability of users DF
Detectability of users data during P	(XI) Undetectability of users data during P
Detectability of users data through AP	(XII) Undetectability of users data through AP
Information disclosure on users DF	(XIII) Confidentiality of users DF
Information disclosure on users during P	(XIV) Confidentiality of users during P
Information disclosure on users through AP	(XV) Confidentiality of users through AP

Table 5. Mapping of the privacy requirements to the Privacy Enhancing Solutions (PES).

PETs			Elementary privacy objectives																
			I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV		
Processing Layer	Secure and outsourced	Homomorphic Encryption																	
		Secure Multiparty Computation	✓	✓		✓	✓		✓	✓					✓	✓			
		Zero-knowledge proofs																	
	Anonymization	Differential privacy																	
		Syntactic	✓	✓		✓	✓		✓	✓		✓	✓						
		Perturbation																	
Communication Layer (Data Flow)	Anonymization	Symmetric and Asymmetric encryption	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓	✓		
		Mixnet Protocol																	
		Tor-related Protocols	✓		✓	✓		✓	✓		✓	✓		✓				✓	
		Random Walk/DHT Protocols																	
		DCnet Protocols																	
	Authentication	Digital Signature																	
		Certificateless signature	✓		✓	✓		✓	✓		✓			✓	✓		✓		
		Quantum-safe Signatures																	
		Homomorphic Signature																	

A number of viewpoints regarding the proposed methodology are worth highlighting. *LINDDUN* at its core provides systematic support for identifying and mitigating threats in software architectures, which we apply to a cyber-physical system in this work. It provides traceability of the privacy assessment process and accountability for GDPR through the systematicity and documentation of each step. Using the mapping table and threat trees as input, it gives a certain level of thoroughness that can be guaranteed. However, the downside is that it's time-consuming and still requires some expertise. First, because each one of the x-mark in Tab.2 must be examined. Second, because we must be vigilant not to overlook any threats to privacy in the realization of the threat tree (Fig. 3).

In our future work, we plan to perform the risk-based prioritization part of this method. It is a risk rating methodology for calculating and evolving and assessing the privacy risks in the CAV misuse case above.

Acknowledgements

This research is funded by the German-French joint collaboration program on “Cybersecurity” funded by the MESRI (via ANR) and BMBF (via VDI/VDE IT) (ANR-20-CYAL-0008-01). In addition, we thank the anonymous reviewers for their valuable comments.

References

- [1] ENISA good practices for security of Smart Cars, November 25, 2019.
- [2] Commission National for Information and Freedom, “CONNECTED VEHICLES AND PERSONAL DATA”, OCTOBER 2017.
- [3] H. Askaripoor, M. Hashemi Farzaneh & A. Knoll. E/E Architecture Synthesis: Challenges and Technologies. Electronics, 11(4), 518, 2022.
- [4] Z. El-Rewini, K. Sadatsharan, D.F. Selvaraj, S. Plathottam, P. Ranganathan. Cybersecurity challenges in vehicular communications, 2020.
- [5] EU General Data Protection Regulation (2016), <http://www.eugdpr.org/>.
- [6] Deng, M., Wuyts, K., Scandariato, R., Preneel, B., Joosen, W.: A privacy threat analysis framework: Supporting the elicitation and fulfillment of privacy requirements. Requirements Engineering 16(1), 2011.
- [7] C. Huang, R. Lu, and K.R. Choo, “Vehicular Fog Computing: Architecture, Use Case, and Security and Forensic Challenges,” no. Nov 2017.
- [8] M.M. Hussain, M.S. Alam, M.M.S. Beg Fog Computing Model for Evolving Smart Transportation Applications Rajkumar Buyya (Ed.), Fog and Edge Computing: Principles and Paradigms, 2019.
- [9] G. S. Andreas and A. L. Opdahl, “Templates for misuse case description,” in Proceedings of the 7 th International Workshop on Requirements Engineering, Foundation for Software Quality, pp. 4–5, 2001.
- [10] P. Akshay, Y.T. Alicia, D. Amrit, B. Mugdha. Privacy Attack for Open Sourced Autonomous Vehicle Data. May 15, 2020.
- [11] J. Ni, K. Zhang, Y. Yu, X. Lin, and X. Shen, “Privacy-preserving smart parking navigation supporting efficient driving guidance retrieval,” IEEE Trans. Veh. Technol., vol. 67, no. 7, pp. 6504–6517, Jul. 2018.
- [12] GM. Garrido, J. Sedlmeir, Ö. Uludag, I.S. Alaoui, A. Luckow, F. Matthes. Revealing the landscape of privacy-enhancing technologies in the context of data markets for the IoT: A systematic literature review. (2021).
- [13] B. Coll-Perales, G. Thandavarayan, M. Sepulcre and J. Gozalvez, “Context-based broadcast acknowledgement for enhanced reliability of cooperative V2X messages”, Proc. Forum Integr. Sustain. Transp. Syst. (FISTS), pp. 393-398, Nov. 2020.
- [14] T. Omitola, B. Waterson, G. Wills, N. Tsakalakis, S. Stalla-Bourdillon, R. Gomer and T. Cherrett. User configurable privacy requirements elicitation in cyber-physical systems, 2022.
- [15] S. WRIGHT. Autonomous cars generate more than 300 TB of data per year, JULY 2, 2021.
- [16] B. Schneier. 1989. Attack Trees. Dr. Dobbs's Journal (December 1989).
- [17] K. Koscher, A. Czeskis, F. Roesner, S. Patel, T. Kohno, S. Checkoway, D. McCoy, B. Kantor, D. Anderson, H. Shacham, et al., Experimental security analysis of a modern automobile, in: Proceedings of the IEEE Symposium on Security and Privacy, IEEE, 2010.
- [18] A. Kovelman. A remote attack on the Bosch Drivelog connector dongle, Argus blog, Argus Cyber Security, <https://argus-sec.com/remote-attack-bosch-drivelog-connector-dongle/> (accessed: 21-07-18).
- [19] S. Checkoway, D. McCoy, B. Kantor, D. Anderson, H. Shacham, S. Savage, K. Koscher, A. Czeskis, F. Roesner, T. Kohno, Comprehensive experimental analyses of automotive attack surfaces, in: Proceedings of the 20th USENIX Conference on Security, 2011.
- [20] V. H. Le, J. D. Hartog, Z. Zannone, “Security and Privacy for Innovative Automotive Applications: A Survey”, Computer Communications, 132, 17–41, 2018.
- [21] K. Ullah, I. Rashid, H. Afzal, W. Iqbal, Y. A. Bangash and H. Abbas, “SS7 vulnerabilities—A survey & implementation of machine learning vs rule based filtering for detection of SS7 network attacks”, IEEE Commun. Surveys Tuts.
- [22] O. Lotfi ben; W. Harold; M. Mohd Murtadha; W. Marko. A survey of security and privacy in connected vehicles. 2015.
- [23] N. Nir, Y. Ran, E. Yuval. USB-based attacks. 2017.
- [24] Z. Cai, A. Wang, W. Zhang, 0-days & mitigations: roadways to exploit and secure connected BMW cars, in: Black Hat USA, 2019.
- [25] X. Sun, F. Richard Yu, Fellow, and P. Zhang, A Survey on Cyber-Security of Connected and Autonomous Vehicles (CAVs), in IEEE.
- [26] I. Rouf, R. Miller, H. Mustafa, T. Taylor, S. Oh, W. Xu, M. Gruteser, W. Trappe, and I. Seskar. Security and privacy vulnerabilities of in-car wireless networks: A tire pressure monitoring system case study. (2010)
- [27] R. Clarke. Privacy impact assessment: Its origins and development. Computer law security review, 25(2), 123-135. 2009.
- [28] A.S. Ahmadian, D. Strüder, V. Riediger & J. Jürjens. Supporting privacy impact assessment by model-based privacy analysis. In Proceedings of the 33rd Annual ACM Symposium on Applied Computing, 2018.
- [29] H. Askaripoor, M. Hashemi Farzaneh & A. Knoll. E/E Architecture Synthesis: Challenges and Technologies. Electronics, 11(4), 518, 2022.
- [30] A. Robles-González, J. Parra-Arnau & J. Forné. A LINDDUN-Based framework for privacy threat analysis on identification and authentication processes. Computers Security, 2020.